

```
1 • CREATE DATABASE bookflow_dbbb;
2 • USE bookflow_dbbb;
3
4 • CREATE TABLE Books (
5     book_id INT PRIMARY KEY,
6     title VARCHAR(100) NOT NULL,
7     isbn VARCHAR(20) UNIQUE,
8     published_year INT CHECK (published_year < 2027)
9 );
10
11 • INSERT INTO Books (book_id, title, isbn, published_year) VALUES
12 (1, 'Gautam', '9780743273565', 1925),
13 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
14 (3, '1984', '9780451524935', 1949);
15 • SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	Gautam	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
* NULL	NULL	NULL	NULL

Books 1 x Apply Revert

#	Time	Action	Message
✓ 1	12:22:41	CREATE DATABASE bookflow_dbbb	1 row(s) affected
✓ 2	12:22:41	USE bookflow_dbbb	0 row(s) affected
✓ 3	12:22:45	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR...	0 row(s) affected
✓ 4	12:23:03	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'Gautam', '9780743273...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 5	12:23:03	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned

```
13 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
14 (3, '1984', '9780451524935', 1949);
15 • SELECT * FROM Books;
16
17 • CREATE TABLE Members (
18     member_id INT PRIMARY KEY,
19     full_name VARCHAR(100),
20     email VARCHAR(100) UNIQUE
21 );
22
23 • INSERT INTO Members (member_id, full_name, email) VALUES
24 (101, 'Gautam', 'laxmanreddy@email.com'),
25 (102, 'Emma Wilson', 'emma.wilson@email.com'),
26 (103, 'Michael Brown', 'michael.brown@email.com');
27 • SELECT * FROM Members;
```

member_id	full_name	email
101	Gautam	reddy@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com
* NULL	NULL	NULL

Members 2 x Apply Revert

#	Time	Action	Message
✓ 4	12:23:03	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'Gautam', '978074327...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 5	12:23:03	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned
✓ 6	12:23:47	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VAR...	0 row(s) affected
✓ 7	12:23:53	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Gautam', 'laxmanreddy...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓ 8	12:23:53	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned

Limit to 1000 rows

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35 FOREIGN KEY (book_id) REFERENCES Books(book_id)
36 );
37
38 • INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
39 (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
40 (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
41 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
42 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
43 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
44 • SELECT * FROM Loans;

```

Result Grid

	loan_id	member_id	book_id	loan_date
▶	1	101	1	2025-01-05
	2	102	2	2025-01-08
	3	103	3	2025-01-10
	4	101	2	2025-02-01
	5	102	1	2025-02-05
	6	103	2	2025-02-12
	7	101	3	2025-03-01
	8	102	3	2025-03-07
	9	103	1	2025-03-15
	10	101	1	2025-04-01
*	NULL	NULL	NULL	NULL

Loans 3 x Apply Revert

Output

Action Output

#	Time	Action	Message
✓	7 12:23:53	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Laxman Reddy', 'laxmanreddy...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	8 12:23:53	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
✓	9 12:24:29	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected
✓	10 12:24:40	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓	11 12:24:41	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned

Limit to 1000 rows

```

41 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
42 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
43 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
44 • SELECT * FROM Loans;
45
46 • SELECT m.full_name AS Member_Name, b.title AS Book_Title
47 FROM Loans l
48 INNER JOIN Members m ON l.member_id = m.member_id
49 INNER JOIN Books b ON l.book_id = b.book_id;
50

```

Result Grid

	Member_Name	Book_Title
▶	Gautam	
	Gautam	To Kill a Mockingbird
	Gautam	1984
	Gautam	Gautam
	Emma Wilson	To Kill a Mockingbird
	Emma Wilson	Gautam
	Emma Wilson	1984
	Michael Brown	1984
	Michael Brown	To Kill a Mockingbird
	Michael Brown	Gautam

Result 4 x Read Only

Output

Action Output

#	Time	Action	Message
✓	8 12:23:53	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned
✓	9 12:24:29	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected
✓	10 12:24:40	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
✓	11 12:24:41	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
✓	12 12:25:19	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Member...	10 row(s) returned

Limit to 1000 rows

```

50
51 • SELECT published_year, COUNT(book_id) AS Total_Books
52 FROM Books
53 GROUP BY published_year
54 ORDER BY published_year;
55
56 • CREATE TABLE Donation_History (
57     donation_id INT PRIMARY KEY,
58     book_id INT,
59     donor_name VARCHAR(100),
60     donation_date DATE
61 );

```

Result Grid

published_year	Total_Books
1925	1
1949	1
1960	1

Result 5 x Read Only

Output

Action Output

#	Time	Action	Message
9	12:24:29	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected
10	12:24:40	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
11	12:24:41	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned
12	12:25:19	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Member...	10 row(s) returned
13	12:25:42	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_ye...	3 row(s) returned

Limit to 1000 rows

```

62 );
63 • START TRANSACTION;
64 • INSERT INTO Books (book_id, title, isbn, published_year)
65 VALUES (4, 'Animal Farm', '9780451526342', 1945);
66 • INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
67 VALUES (1, 4, 'Gautam' CURDATE());
68 • COMMIT;
69
70 • CREATE INDEX idx_books_isbn ON Books(isbn);
71 • SELECT * FROM Books WHERE isbn = '9780451524935';

```

Result Grid

book_id	title	isbn	published_year
3	1984	9780451524935	1949
* NULL	NULL	NULL	NULL

Books 6 x Apply Revert

Output

Action Output

#	Time	Action	Message
16	12:26:20	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '97804515263...	1 row(s) affected
17	12:26:24	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4,...	1 row(s) affected
18	12:26:27	COMMIT	0 row(s) affected
19	12:26:35	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0
20	12:26:41	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned